Why Eight Pi Is Critical:
Energy, Virial, and Stationary Profiles in Planar Keller–Segel Flow

HCS-C363 source-local reconstruction

4 September 2026

Abstract

For the parabolic–elliptic Keller–Segel equation on the plane, we derive the critical mass 8π from two independent exact mechanisms. A mass-preserving dilation changes free energy by $2M(1 - M/(8\pi))\log\lambda$, while Newtonian-kernel symmetrization gives the finite-moment virial law $I' = 4M(1 - M/(8\pi))$. The latter forces every supercritical classical finite-moment solution to leave that regime no later than an explicit time. At critical mass we verify the full translated and dilated stationary family, then compute its logarithmically infinite second moment; this hypothesis check removes an apparent conflict with the zero virial slope. We also derive the radial cumulative-mass equation and its critical equilibrium. Exact symbolic, rational, replay, and hostile receipts audit coefficients but do not replace the PDE proof. No literature-priority, weak-continuation, or arithmetic-target claim is made.

Revision certificate. Round zero energy scaling and critical mass closure.

1 Equation, hypotheses, and main result

We use the Newtonian convention

$$\rho_t = \Delta\rho - \nabla\cdot(\rho\nabla c), \quad -\Delta c = \rho, \quad c(x) = -\frac{1}{2\pi}\int_{\mathbb{R}^2} \rho(y)\log|x-y|\,dy. \tag{1}$$

Every assertion is made on an open interval where ρ is nonnegative, C^1 in time and C^2 in space, has finite mass, the logarithmic convolution defining c exists, and all displayed fluxes, derivatives, and cutoff limits are integrable. Barycenter conservation requires a finite first moment; the virial law requires a finite second moment. The free-energy identity further requires finite entropy and interaction energy, strict positivity when $M > 0$, and finite dissipation. The zero solution is treated separately without invoking $\log\rho$. These assertion-specific hypotheses are part of the theorem.

Theorem 1 (Critical-mass atlas)**.** *Let $M = \int\rho$ and, when finite, $B = \int x\rho$ and $I = \int |x|^2\rho$. Along* (1)*,*

$$M' = 0, \quad B' = 0, \tag{2}$$

$$\frac{d}{dt}\mathcal{F}[\rho] = -\int \rho\,|\nabla(\log\rho - c)|^2, \quad \mathcal{F}[\rho] = \int \rho\log\rho - \frac{1}{2}\int \rho c, \tag{3}$$

$$I' = 4M\left(1 - \frac{M}{8\pi}\right). \tag{4}$$

For $\rho_\lambda(x) = \lambda^2\rho(\lambda x)$,

$$\mathcal{F}[\rho_\lambda] - \mathcal{F}[\rho] = 2M\left(1 - \frac{M}{8\pi}\right)\log\lambda. \tag{5}$$

If $M > 8\pi$, a classical finite-second-moment solution cannot persist beyond

$$T_* = \frac{2\pi I(0)}{M(M-8\pi)}. \tag{6}$$

At $M = 8\pi$, every $\lambda > 0$ and $a \in \mathbb{R}^2$ give a stationary pair

$$\rho_{\lambda,a}(x) = \frac{8\lambda^2}{(\lambda^2 + |x-a|^2)^2}, \quad c_{\lambda,a}(x) = -2\log(\lambda^2 + |x-a|^2) + C. \tag{7}$$

It has mass 8π and infinite second moment.

2 Conservation, energy, and scaling

Integrating (1) proves mass conservation. For the barycenter, diffusion integrates to zero and

$$\int \rho(x)\nabla c(x)\,dx = -\frac{1}{2\pi}\iint \rho(x)\rho(y)\frac{x-y}{|x-y|^2}\,dx\,dy = 0,$$

because exchanging x, y negates the integrand. This proves (2).

The first equation in (1) is

$$\rho_t = \nabla\cdot\{\rho\nabla(\log\rho - c)\}. \tag{8}$$

The first variation of $\mathcal{F}$ is $\log\rho + 1 - c$; the constant contributes nothing because mass is conserved. Integration by parts gives (3).

Dilation contributes $2M\log\lambda$ to $\int \rho\log\rho$. Since

$$-\frac{1}{2}\int \rho c = \frac{1}{4\pi}\iint \rho(x)\rho(y)\log|x-y|\,dx\,dy,$$

changing variables in the double integral contributes $-M^2\log\lambda/(4\pi)$. Their sum is (5). Hence free energy increases along concentration dilations below 8π, is scale invariant at 8π, and decreases along concentration dilations above 8π. This scaling statement alone is not an existence or blow-up theorem.